Xiangyu Wang

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Education

University of Illinois Urbana-Champaign Fall 2023 - present
Ph.D. in Mathematics, in progress
Advisor: Xiaochun Li

Research Interests

Harmonic analysis (Kakeya-type problems, Fourier restriction, decoupling, and local smoothing); additive combinatorics; analytic number theory.

Publications and Preprints

1. *Restriction Estimates Using Multilinear Kakeya* arXiv:2610.17752
   For sufficiently large dimensions, our approach yields better exponents than those obtained by a direct application of polynomial partitioning.
2. *Cone Restriction Estimates under Half-Scale Dimensional Conditions* arXiv:2610.16294
   Building on the framework of Du--Li--Wang--Zhang, we incorporate the broad restriction estimate of Ou--Wang and a localized multilinear estimate obtained from cyclic bilinear pairing and Kakeya summation of packet energies.
3. *Wave Packet Density in Local Smoothing: Refinements and Limitations* arXiv:2609.34675
   This work refines the wave packet density framework to improve local smoothing estimates for wave equations in all even spatial dimensions $n \geq 4$ including $p > 2.8862\ldots$ when $n = 4$. It also constructs an explicit Euclidean counterexample that reveals limitations of the existing framework.
4. *A Weighted Restriction Estimate in* $\mathbb{R}^2$ arXiv:2609.12587
   with Zhenbin Cao, Feilong Guo, and Junfeng Li.
5. *On Oscillatory Integral Operators satisfying the Cinematic Curvature Condition* arXiv:2512.22516
6. *On Cone Restriction Estimates in Higher Dimensions* arXiv:2411.02961
   Accepted by Journal of Fourier Analysis and Applications.
7. *On Integers whose Friable Component is Large* arXiv:2609.37507
   with Gérald Tenenbaum.
8. *A Short Proof of a Theorem of Pan and Zhang* arXiv:2609.37508

Teaching Experience

University of Illinois Urbana-Champaign
Scheduled to lead discussion sections in Spring 2027.
Mathematics Tutor (Provided homework help and answered students' questions): MATH 241 (Calculus III).
Grader: MATH 512 (Algebraic Geometry), MATH 444 (Elementary Real Analysis), and MATH 448 (Complex Variables).